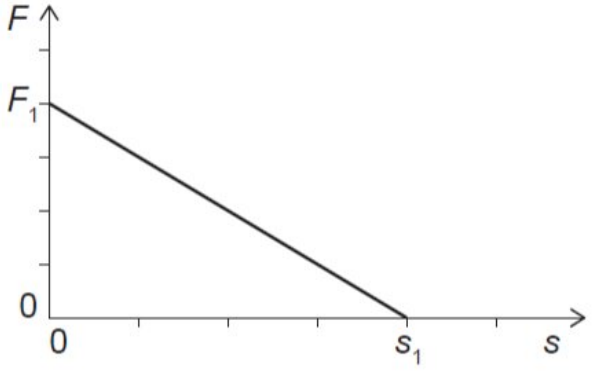
4 A net force F acts on an object of mass m that is initially at rest. The object moves in a straight line. The variation of F with the distance s is shown.



What is the speed of the object at the distance s?

A $\sqrt{\frac{F_1 s_1}{2m}}$ **B** $\sqrt{\frac{F_1 s_1}{m}}$ **C** $\sqrt{\frac{2F_1 s_1}{m}}$ **D** $\sqrt{\frac{4F_1 s_1}{m}}$